

Environment / Stream

Event stream
 (u_t, v_t^*, t)

Performative Feedback

Update backbone
& graph stats

Policy + Logging

Candidate Set
 $C_t = \{v_t^*\} \cup \text{Negatives}$

Backbone
Scoring
 $f_t \rightarrow \hat{p}$

Decision-
layer logit
shift $\delta(a)$

Apply
 r_0 -calibrator

Coverage
 $\rightarrow \epsilon_{\text{eff}} + \text{mask}$

Policy π_t
(topk-stoch/
 ϵ_{eff} -greedy)

Logged
propensities
 $\hat{e}_t$

Observe
outcome y_t

Deployed
hit?

YES

NO

No update
(bandit)

Estimation

(Cross-fit + GA-DR)

DR ingest

Online cross-fit
 $\hat{u}_0, \hat{u}_1$

GA-DR
buffer

Auditing + Control

Update
Multicalibrator
(every K steps)

Fairness metrics
 $g\text{TE}_{\text{gap}}, g\text{Cal}_{\text{max}}$

Residual-Ol auditor
Residual-Ol bounds

PI dual update
 $\lambda_{\text{TE}}, \lambda_{\text{Cal}}, \lambda_{\text{Min}}$